

3. Replicable

Both the asset owner and the funds manager should be able to replicate the benchmark. Martingale is certainly able to replicate the returns of the Russell 1000 benchmark because it bases its strategy on the Russell 1000 universe. GM Asset Management, the client, is also able to replicate the Russell 1000, either by internally trading it or by employing a Russell 1000 index fund provider. Thus, both Martingale and GM face a common, low-cost option.

Certain benchmarks can't be replicated by the asset owner because they are beyond the asset owner's expertise. Such nonreplicable benchmarks are not viable choices and make it difficult or impossible to measure how much value a portfolio manager has added because the benchmark itself cannot be achieved by the asset owner. There are some benchmarks, like absolute return benchmarks, that can't even be replicated by the fund manager. In these cases, the fund manager is disadvantaged because she may not even be able to generate the benchmark in the first place.

4. Adjusted for risk

Sadly, most benchmarks used in the money management business are not risk adjusted.

Taking the Russell 1000 as the benchmark assumes the beta of Jacques's strategy is one, but the actual beta of the volatility strategy is 0.73. This risk adjustment makes a big difference in the alpha; with the true beta of 0.73, the alpha of the low volatility strategy is 3.44% per year compared to 1.50% when the beta is one.

When we compute Martingale's beta, we assume that the investor can construct the benchmark portfolio ($r_t^{bmk} = 0.27r_t^f + 0.73r_t^{R1000}$) of T-bills and the Russell 1000 and rebalance it every month. GM Asset Management can easily do this, but some risk-adjusted benchmarks are beyond the reach of less sophisticated clients. We also estimate the beta using data from the whole sample. That is, the beta of 0.73 is the beta generated by Martingale after the fact—at the beginning of the sample, it was not obvious that Martingale would actually trade to generate a beta of 0.73.

And even though Martingale endorses the Russell 1000 as its benchmark, beta-adjusted or not, what if this is the wrong adjustment for risk? We know that there are more risk factors than just the equity market (see chapter 7): dynamic factors like value-growth, momentum, credit, and volatility risk also exist. Perhaps we should adjust for some of these as well? In section 3 we extend risk adjustments to multiple sources of risk.

2.3 CREATING ALPHA

A portfolio manager creates alpha relative to a benchmark by making bets that deviate from that benchmark. The more successful these bets are, the higher the alpha.